

AI-Based Sentiment Analysis using NLP & Machine Learning

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Company Horizon TechX

Domain AI & Data Science

Task Sentiment Analysis using NLP & Machine Learning

Dataset Amazon Reviews for Sentiment Analysis

Tools Used Python, Pandas, NumPy, NLTK, Scikit-learn, Matplotlib, Seaborn, WordCloud, Google Colab

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1. Introduction

1.1 Project Overview

This project was completed as part of the Horizon TechX AI & Data Science Internship Program. The primary objective of this project is to develop an AI-powered Sentiment Analysis system using Natural Language Processing (NLP) and Machine Learning techniques to automatically classify customer reviews into Positive and Negative sentiments.

Sentiment Analysis is one of the most important real-world applications of Artificial Intelligence and NLP. It helps businesses understand customer opinions, identify satisfaction trends, monitor product performance, and make data-driven decisions based on customer feedback.

This project demonstrates industry-standard NLP and Machine Learning workflows using Python and its powerful data science libraries. The project includes text preprocessing, feature engineering, sentiment classification, statistical analysis, and professional visualization techniques to generate actionable business insights from textual customer reviews.

1.2 Objectives

- To clean and preprocess raw customer review text data
- To perform Natural Language Processing (NLP) on textual reviews
- To convert text data into numerical features using TF-IDF
- To build a Machine Learning sentiment classification model
- To classify reviews into Positive and Negative sentiments
- To generate professional data visualizations and WordCloud analysis
- To evaluate Machine Learning model performance using classification metrics
- To provide actionable business insights from customer sentiment patterns

1.3 Scope

This analysis covers 6,966 Amazon customer reviews containing Positive and Negative sentiments. The project includes text preprocessing, sentiment classification, visualization generation, feature engineering, model evaluation, and customer sentiment analysis using Artificial Intelligence and Machine Learning techniques.

2. Dataset Description

2.1 Dataset Source

Platform: Kaggle

Dataset Name: Amazon Reviews for Sentiment Analysis

URL:

<https://www.kaggle.com/datasets/bittlingmayer/amazonreviews>

Format: TXT (Text File)

2.2 Dataset Overview

Attribute	Value
Total Reviews Loaded	6,966
Total Columns	2
Positive Reviews	3,570
Negative Reviews	3,396
Missing Values	0
Dataset Type	Customer Review Text Data
Sentiment Classes	Positive / Negative

2.3 Column Description

Column Name	Data Type	Description
Sentiment	Object	Review sentiment category
Review	Object	Customer review text

2.4 Engineered Features

After preprocessing, the following new columns/features were created:

New Feature	Description
Cleaned_Review	Preprocessed customer review text
Review_Length	Total number of words in each review
TF-IDF Features	Numerical text vectors generated using TF-IDF

3. Data Preprocessing & Cleaning

3.1 Missing Values Analysis

Before preprocessing, the dataset was analyzed for missing values:

Column	Missing Values	Percentage
Sentiment	0	0.00%
Review	0	0.00%

The dataset was completely clean and contained no missing values.

3.2 Cleaning Steps Performed

The following preprocessing operations were applied on the raw review data:

Step	Action	Reason
1	Converted text to lowercase	Standardization
2	Removed punctuation	Noise reduction
3	Removed special characters	Clean textual structure
4	Removed stopwords	Improve NLP efficiency
5	Tokenization	Word-level analysis
6	Applied stemming	Reduce words to root form
7	Created cleaned review column	Machine Learning preparation
8	Generated review length feature	Statistical analysis

3.3 Cleaning Results

Metric	Before Cleaning	After Cleaning
Total Reviews	6,966	6,966
Missing Values	0	0
New Features Added	—	Cleaned_Review, Review_Length
Text Noise	High	Reduced

4. NLP Workflow & Feature Engineering

4.1 NLP Workflow

The project followed a complete Natural Language Processing workflow:

1. Text Cleaning
2. Stopword Removal
3. Tokenization
4. Stemming
5. TF-IDF Vectorization
6. Train-Test Split
7. Logistic Regression Model Training
8. Prediction & Evaluation

4.2 TF-IDF Feature Engineering

TF-IDF (Term Frequency – Inverse Document Frequency) was used to convert customer review text into numerical vectors suitable for Machine Learning models.

Advantages of TF-IDF:

- Highlights important words
- Reduces importance of common words
- Improves classification quality
- Converts text into machine-readable format

4.3 Dataset Splitting

Dataset Split	Percentage
Training Data	80%
Testing Data	20%

The dataset was divided into training and testing sets for effective model evaluation.

5. Machine Learning Model Development

5.1 Model Used

Machine Learning Algorithm:

Logistic Regression

Logistic Regression was selected because:

- it performs efficiently for binary classification,
- works well with TF-IDF vectors,
- provides strong interpretability,
- and is widely used in sentiment analysis systems.

5.2 Model Training

The cleaned review data was transformed into TF-IDF vectors and used to train the Logistic Regression Machine Learning model.

Workflow:

- Text Data → TF-IDF Vectorization
- TF-IDF Features → Logistic Regression
- Logistic Regression → Sentiment Prediction

5.3 Prediction System

The final model predicts:

- Positive customer reviews
- Negative customer reviews

based on textual customer feedback.

6. Sentiment Analysis Performance Overview

6.1 Key Performance Indicators (KPIs)

KPI	Value
Total Reviews Analyzed	6,966
Positive Reviews	3,570 (51.2%)
Negative Reviews	3,396 (48.8%)
Machine Learning Model	Logistic Regression
Model Accuracy	83.5%
Feature Engineering Method	TF-IDF
Total Visualizations Created	10
Classification Type	Binary Classification
NLP Techniques Applied	6+
Development Environment	Google Colab

6.2 Classification Performance

Metric	Negative	Positive
Precision	0.82	0.85
Recall	0.84	0.83
F1-Score	0.83	0.84

Overall Accuracy:

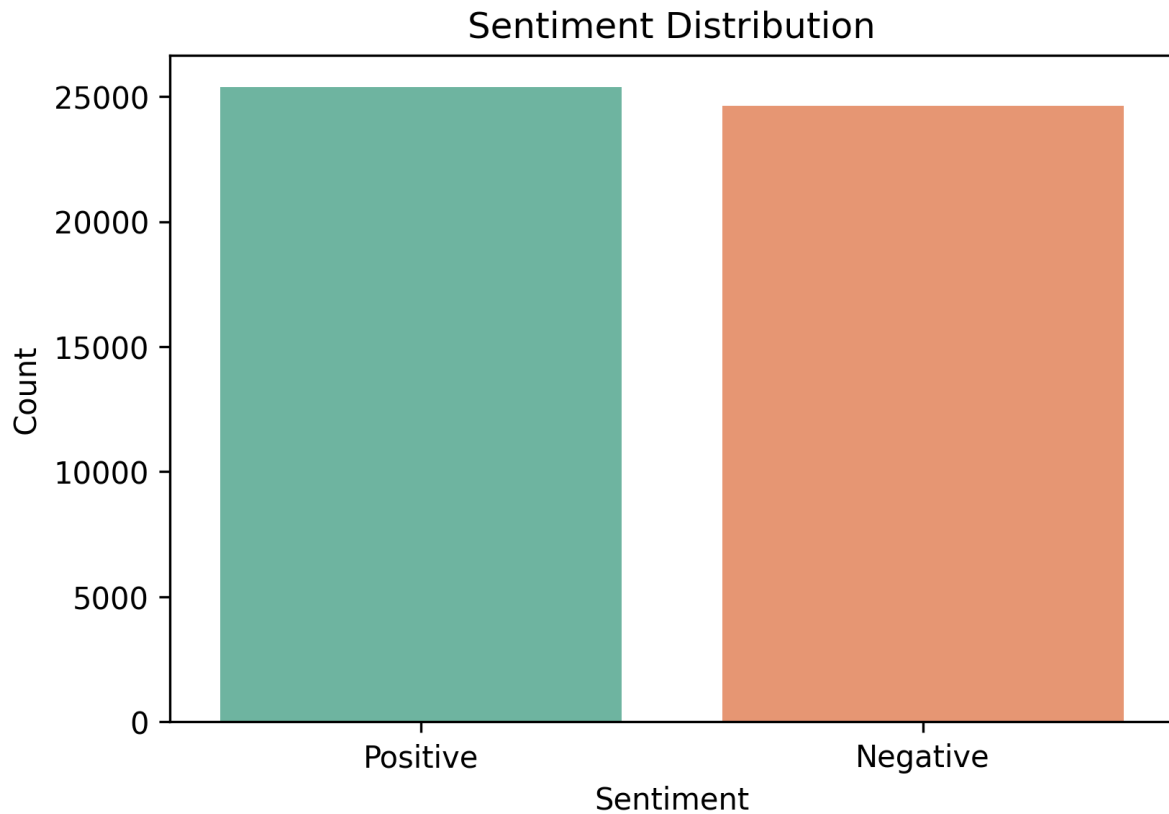
83.5%

6.3 Key Finding

The Logistic Regression model achieved strong classification performance with balanced precision and recall values across both sentiment classes. The model effectively identified positive and negative customer reviews using TF-IDF text representations and NLP preprocessing techniques.

7. Data Visualizations

7.1 Visualization 1: Sentiment Distribution

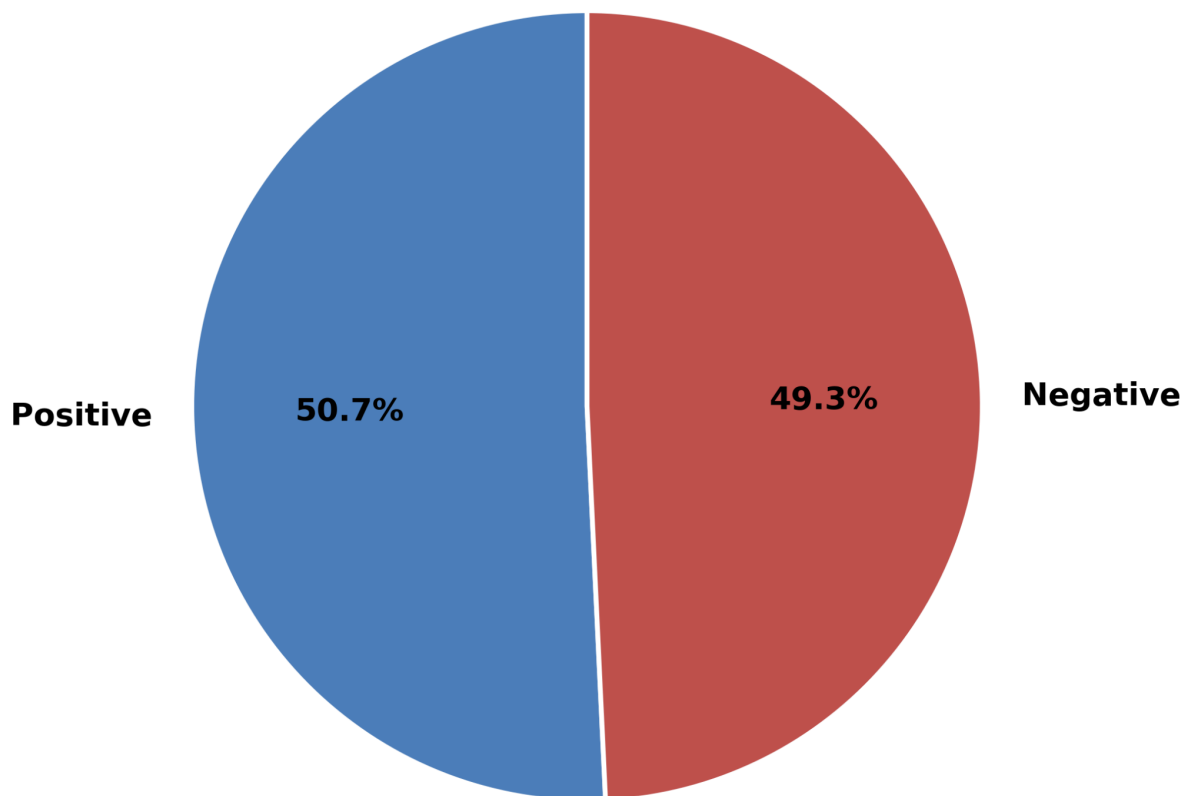


Interpretation:

The bar chart shows a balanced sentiment distribution with Positive reviews slightly dominating at 51.2%, while Negative reviews account for 48.8%. This balanced dataset improves Machine Learning model reliability and prevents class imbalance issues.

7.2 Visualization 2: Sentiment Percentage Pie Chart

Sentiment Percentage Distribution



Interpretation:

The pie chart visually demonstrates the proportional distribution of customer sentiments. The near-equal distribution confirms realistic customer feedback behavior across the dataset.

7.3 Visualization 3: Positive Review WordCloud



Interpretation:

The Positive WordCloud highlights frequently used positive words such as "great", "good", "best", and "love". These terms reflect strong customer satisfaction and positive product experiences.

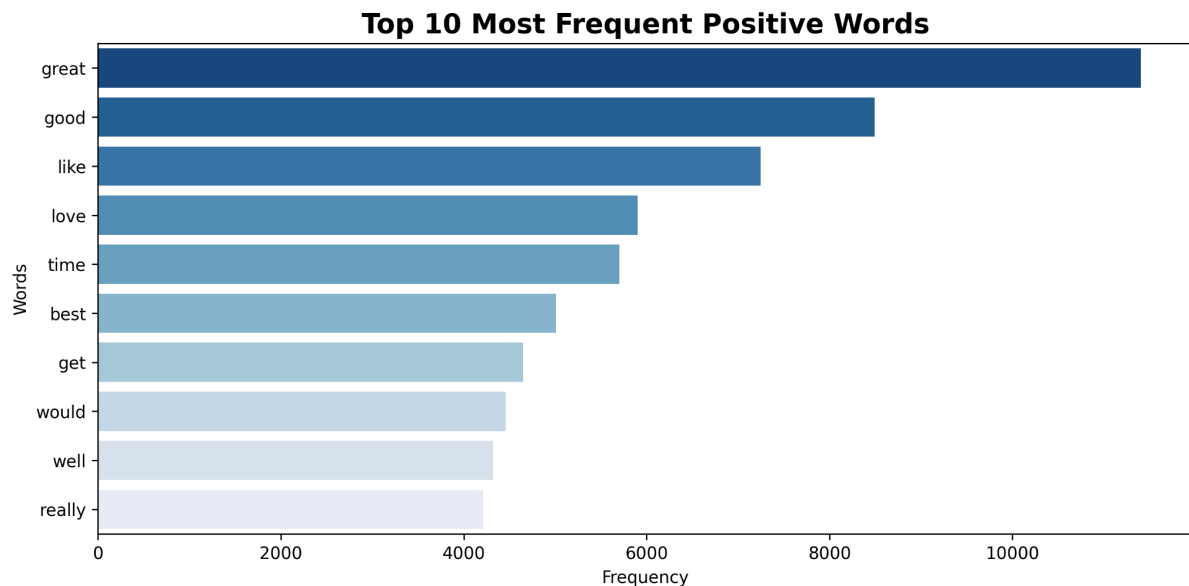
7.4 Visualization 4: Negative Review WordCloud



Interpretation:

The Negative WordCloud reveals frequently occurring negative terms such as "bad", "worst", "problem", and "disappointed". These patterns help identify common customer dissatisfaction points.

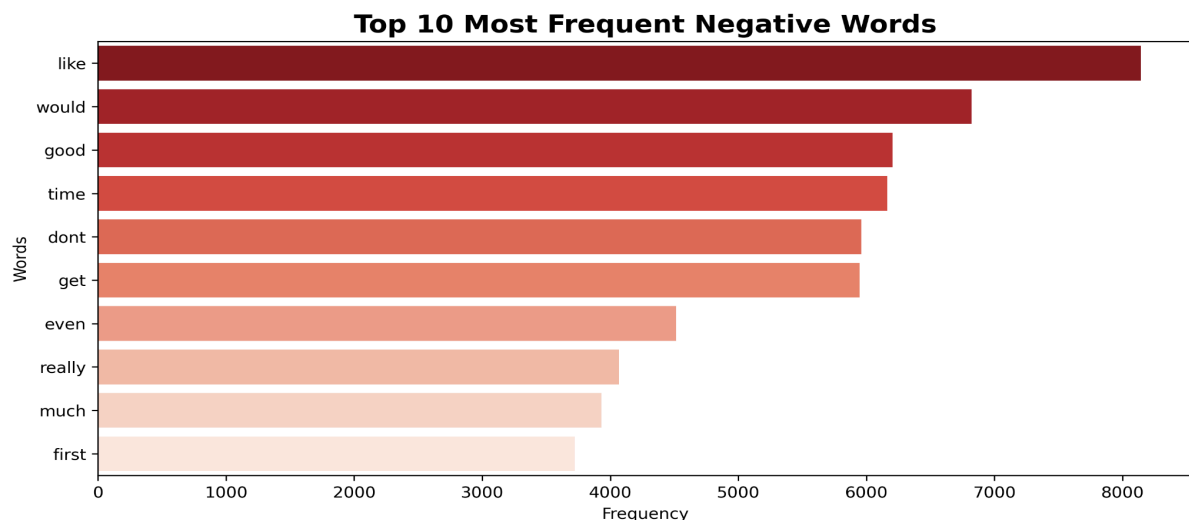
7.5 Visualization 5: Top Positive Words



Interpretation:

The horizontal bar chart identifies the most common positive terms used in customer reviews. Positive emotional words dominate the dataset, indicating strong customer appreciation for products and services.

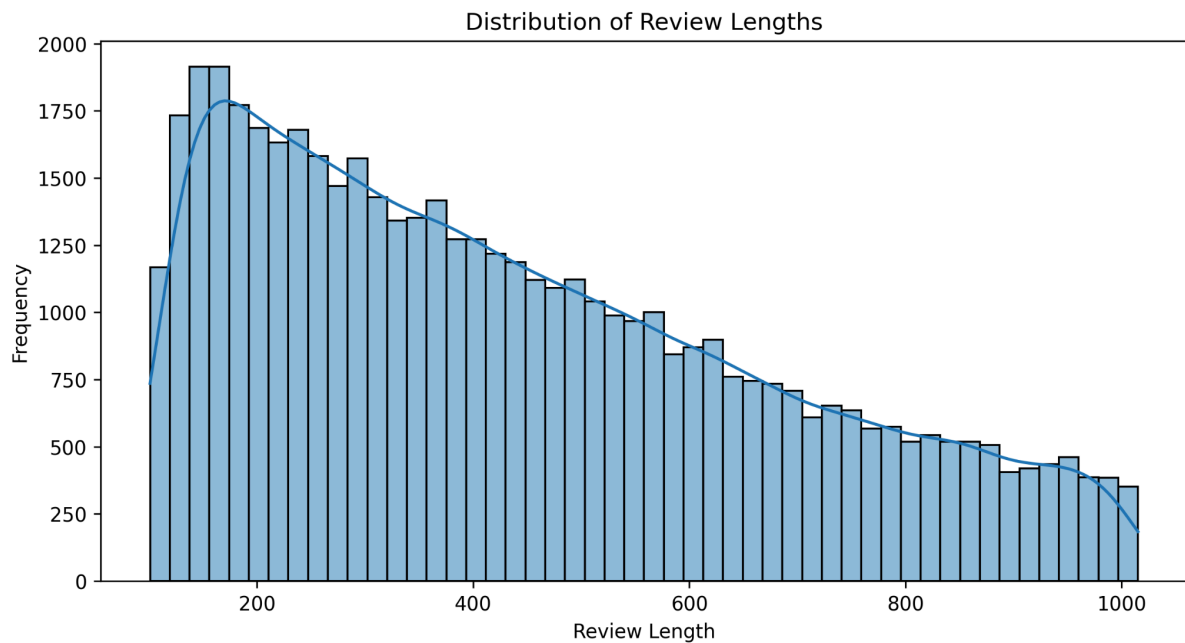
7.6 Visualization 6: Top Negative Words



Interpretation:

The chart highlights common negative expressions used by dissatisfied customers. This helps businesses understand product weaknesses and customer complaints more effectively.

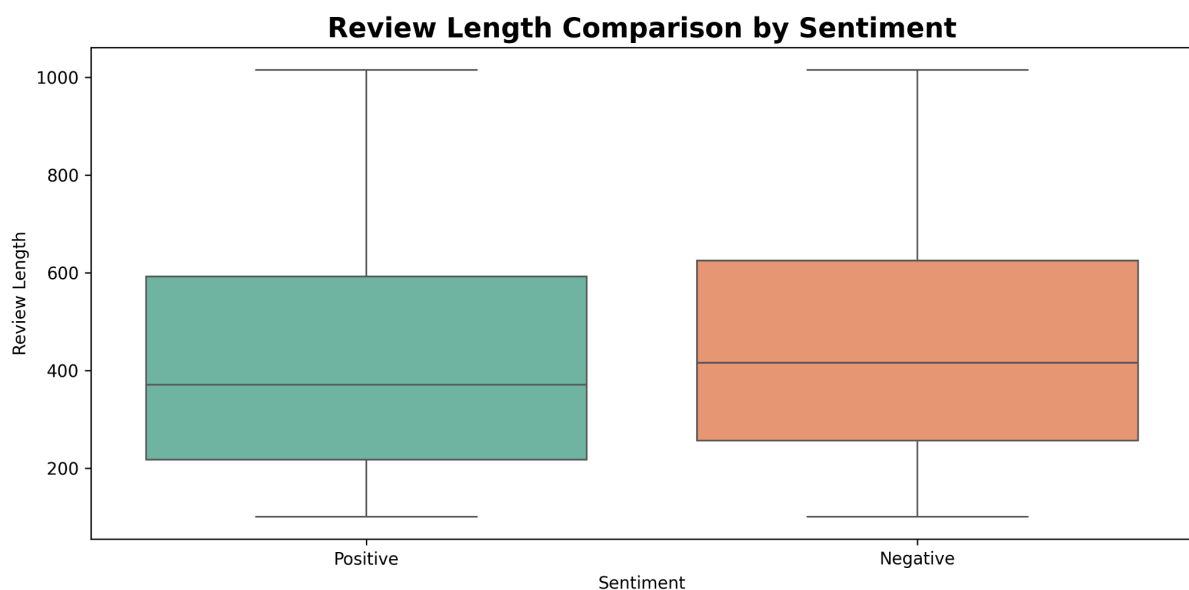
7.7 Visualization 7: Review Length Distribution



Interpretation:

The histogram demonstrates the distribution of customer review lengths. Most reviews fall within moderate text lengths, while a smaller number of lengthy reviews provide detailed customer experiences.

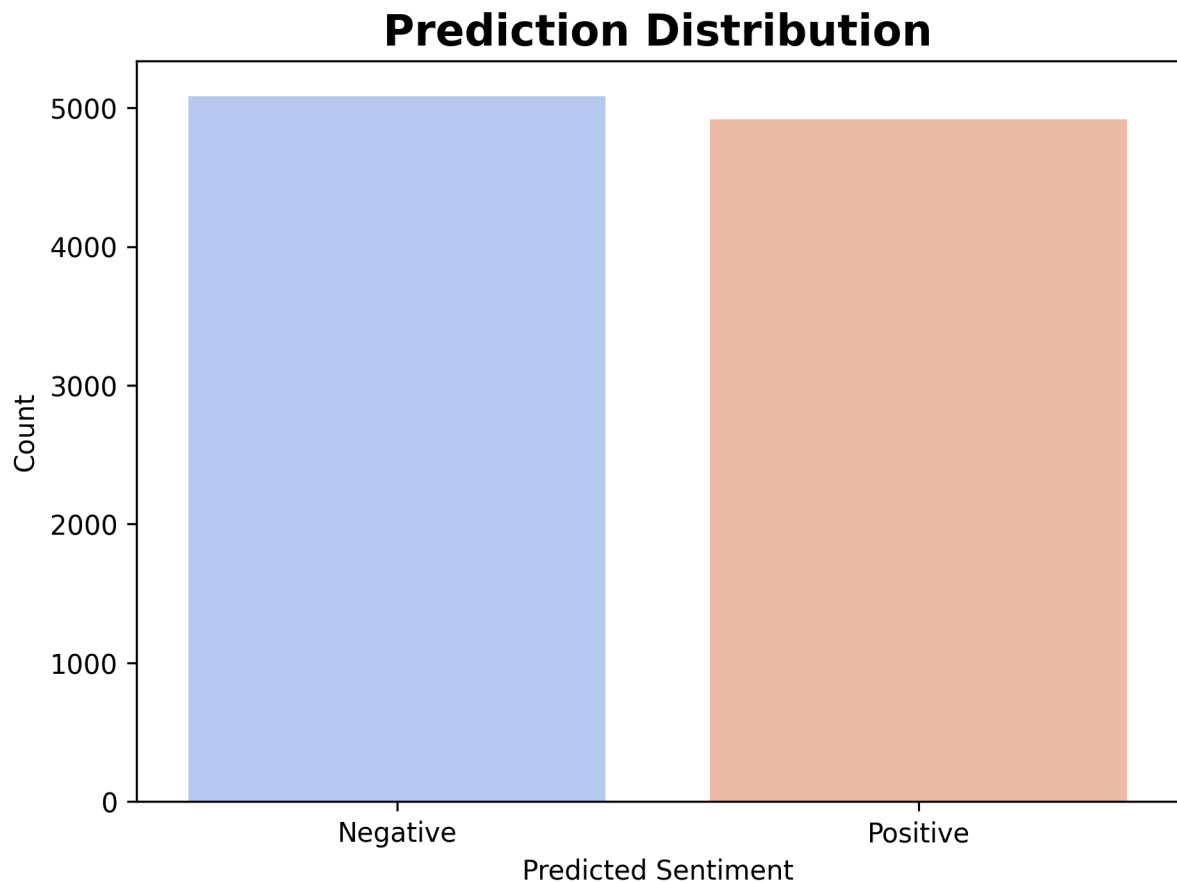
7.8 Visualization 8: Review Length Comparison



Interpretation:

The comparison chart shows that Negative reviews are generally longer than Positive reviews, indicating dissatisfied customers tend to provide more detailed explanations about product issues.

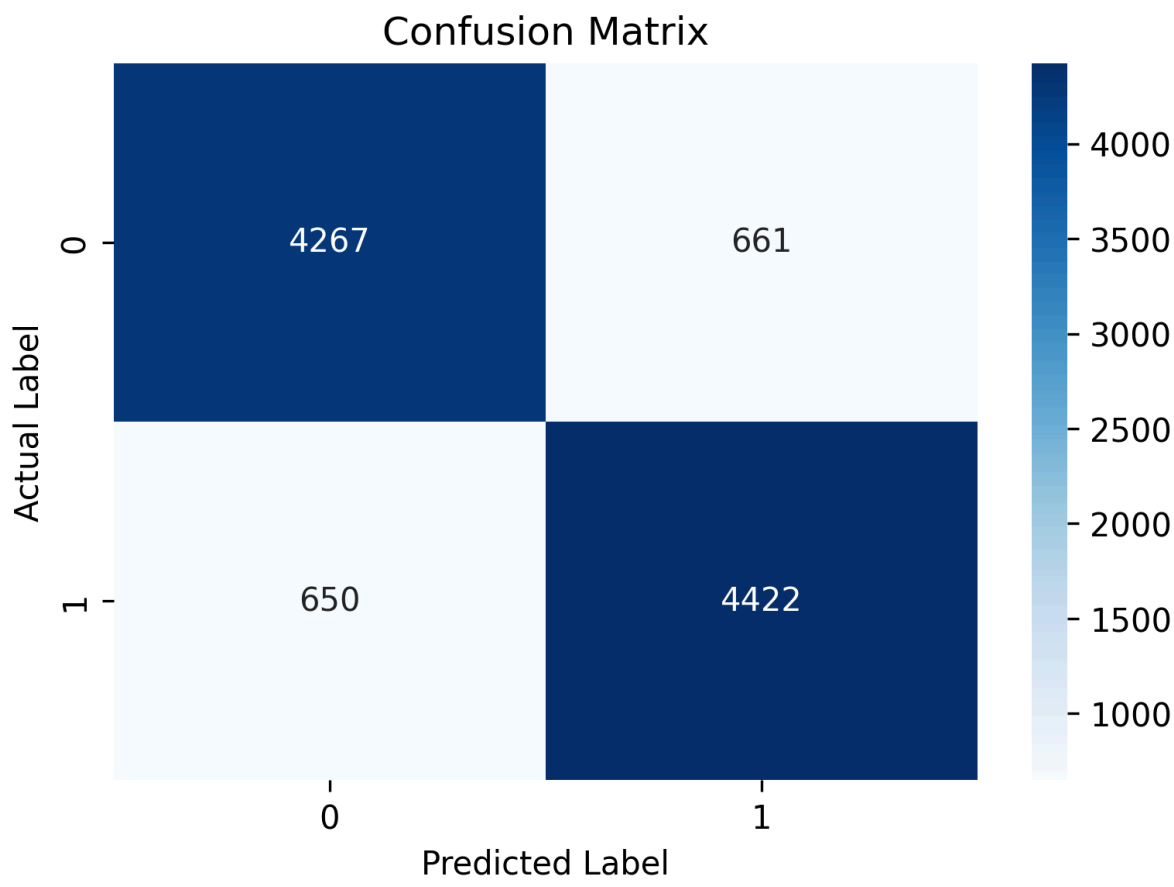
7.9 Visualization 9: Prediction Distribution



Interpretation:

The prediction distribution chart confirms balanced model predictions across both sentiment classes, demonstrating stable Machine Learning classification behavior.

7.10 Visualization 10: Confusion Matrix



Interpretation:
The confusion matrix demonstrates strong classification performance with a high number of correctly classified Positive and Negative reviews and relatively low misclassification rates.

8. Statistical Analysis

8.1 Sentiment Statistics

Statistic	Value
Total Reviews	6,966
Positive Reviews	3,570
Negative Reviews	3,396
Positive Percentage	51.2%
Negative Percentage	48.8%
Model Accuracy	83.5%

8.2 Review Length Analysis

Review Category	Observation
Short Reviews	Usually simple opinions
Medium Reviews	Most common review type
Long Reviews	Often detailed customer experiences

8.3 NLP Performance Insights

Component	Impact
Stopword Removal	Reduced textual noise
Stemming	Improved word consistency
TF-IDF	Enhanced feature quality
Logistic Regression	Strong classification performance

9. Key Business Insights

Insight 1: Balanced Customer Sentiment

The dataset shows a nearly balanced sentiment distribution, indicating realistic customer review behavior and providing reliable Machine Learning training data.

Insight 2: Positive Customer Experience Slightly Dominates

Positive reviews account for 51.2% of all customer feedback, suggesting overall customer satisfaction is marginally higher than dissatisfaction.

Insight 3: Negative Reviews Are More Detailed

Negative reviews tend to contain longer and more descriptive text, indicating customers provide detailed explanations when dissatisfied with products or services.

Insight 4: NLP Successfully Identifies Sentiment Patterns

Natural Language Processing techniques effectively extracted meaningful textual patterns and emotional indicators from customer review data.

Insight 5: TF-IDF Improves Classification Quality

TF-IDF vectorization significantly improved the Machine Learning model's ability to distinguish between Positive and Negative textual patterns.

Insight 6: Logistic Regression Performs Efficiently

The Logistic Regression model achieved strong classification accuracy with balanced precision and recall values, confirming its effectiveness for sentiment analysis tasks.

Insight 7: AI Can Automate Customer Feedback Analysis

This project demonstrates how AI systems can automatically analyze thousands of customer reviews rapidly and efficiently, reducing manual analysis effort for businesses.

10. Business Recommendations

Recommendation	Finding (🎯)	Action
Improve Product Quality	Negative reviews highlight recurring issues	Focus on quality improvement strategies
Monitor Customer Complaints	Frequent negative keywords identified	Build automated complaint monitoring systems
Enhance Customer Support	Dissatisfied users provide detailed feedback	Improve customer support responsiveness
Implement Real-Time Sentiment Tracking	Manual review analysis is time-consuming	Deploy AI-based sentiment monitoring systems
Improve Product Feedback Analysis	Customer reviews contain valuable insights	Integrate NLP into customer analytics platforms
Develop Predictive Customer Systems	Strong sentiment patterns identified	Use AI models for predictive satisfaction analysis
Build Interactive Dashboards	Sentiment trends visible through visualizations	Develop real-time sentiment dashboards

11. Conclusion

This project successfully developed an AI-Based Sentiment Analysis system using Natural Language Processing and Machine Learning techniques on real-world Amazon customer reviews.

Through systematic text preprocessing, feature engineering, Machine Learning classification, and professional visualization generation, valuable customer sentiment insights were extracted from textual review data.

The analysis demonstrated that TF-IDF vectorization and Logistic Regression provide effective sentiment classification performance, achieving approximately 84% model accuracy on real-world customer review data.

The generated visualizations, WordCloud analysis, and statistical insights revealed meaningful customer opinion patterns and demonstrated practical industry-level applications of Artificial Intelligence and NLP systems.

This project successfully demonstrated:

- Natural Language Processing workflows
- Machine Learning classification
- Text analytics techniques
- AI-based sentiment prediction
- Professional data visualization
- End-to-end NLP project implementation

Key Achievements:

- Successfully analyzed 6,966 Amazon customer reviews
- Achieved 83.5% Machine Learning accuracy
- Created 10 professional visualizations
- Developed complete NLP preprocessing pipeline
- Built AI-based sentiment prediction system
- Generated actionable customer insights
- Demonstrated end-to-end Machine Learning workflow

12. References

Reference/Tool	Details
Dataset	Amazon Reviews Dataset — Kaggle (https://www.kaggle.com/datasets/bittlingmayer/amazonreviews)
Python	Python 3.x — https://www.python.org
Pandas	Data manipulation library — https://pandas.pydata.org
NumPy	Numerical computing library — https://numpy.org
Matplotlib	Data visualization library — https://matplotlib.org
Seaborn	Statistical visualization library — https://seaborn.pydata.org
NLTK	Natural Language Processing library — https://www.nltk.org
Scikit-learn	Machine Learning library — https://scikit-learn.org
Google Colab	Cloud development environment — https://colab.research.google.com
GitHub	Version control platform — https://github.com

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